

MONGODB Files

```
mongosh mongodb://127.0.0.1:27021/
base. Read and write access to data and configuration is unrestricted
-----

test> show databases
admin      40.00 KiB
config     36.00 KiB
local      80.00 KiB
people     208.00 KiB
test>
| use people
|
switched to db people
people> db.people.insertOne({name:"Bob", age:32, gender:"Male"})
{
  acknowledged: true,
  insertedId: ObjectId('6aa42f03f99287b5cee55458')
}
people> db.people.insertOne({name:"Raju", age:45, gender:"Male"})
{
  acknowledged: true,
  insertedId: ObjectId('6aa42f0ef99287b5cee55459')
}
people> show dbs
|
admin      40.00 KiB
config     36.00 KiB
local      80.00 KiB
people     248.00 KiB
people> show collections
hobbies
people
person
persons
samples
people> db.people.drop()
true
people>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
people> db.dropDatabase()
{ ok: 1, dropped: 'people' }
people> |
```

```

people> db.person.insertOne({name:"Bob", age:32, gender:"Male"})
{
  acknowledged: true,
  insertedId: ObjectId('6aa43145f99287b5cee5545a')
}
people> db.person.insertMany([{name:"Zen", age:54, gender:"Male"}, {name:"Catherine", age:34, gender:"Female"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aa4316bf99287b5cee5545b'),
    '1': ObjectId('6aa4316bf99287b5cee5545c')
  }
}
people> combination of both insertOne() & insertMany()

```

```

people>
| db.samples.find()
| db.samples.find().pretty()
| db.samples.find().count()
0
people> db.samples.findOne() // will give first document
|
null
people> db.samples.updateOne({_id: ObjectId("5f05788738510f90eb50a889")},{$set:{phone:248485685}})
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
people> db.samples.updateMany({gender:"Female"},{$set:{isFemale:true}})
|
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}

```

```
MongoInvalidArgumentError: Update document requires atomic operators
people> db.samples.updateOne(
|   {_id: ObjectId("5f05788738510f90eb50a889")},
|   {$set:{phone:248485685}}
| )
|
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 0,
|   modifiedCount: 0,
|   upsertedCount: 0
| }
people> db.samples.update({_id: ObjectId("5f05788738510f90eb50a889")},{$set:
| {phone:248485685}})
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 0,
|   modifiedCount: 0,
|   upsertedCount: 0
| }
people> db.samples.update({_id: ObjectId("5f05788738510f90eb50a889")},{$set:
| {phone:248485685}})
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 0,
|   modifiedCount: 0,
|   upsertedCount: 0
| }
people> db.samples.updateOne({_id: ObjectId("5f05788738510f90eb50a889")},{$s
| $set:{phone:248485685}})
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 0,
|   modifiedCount: 0,
|   upsertedCount: 0
| }
people> |
```

```

    modifiedCount: 0,
    upsertedCount: 0

people> db.samples.update({_id: ObjectId("5f05788738510f90eb50a889")}, {$set:
phone:248485685}})

    acknowledged: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0

people> db.samples.updateOne({_id: ObjectId("5f05788738510f90eb50a889")}, {$s
set:{phone:248485685}})

    acknowledged: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0

people> db.samples.replaceOne({_id: ObjectId("5f05788738510f90eb50a889")}, {n
ame:"John", age:42})

    acknowledged: true,
    insertedId: null,
    matchedCount: 0,
    modifiedCount: 0,
    upsertedCount: 0

people> db.person.deleteOne({name:"John"})
    acknowledged: true, deletedCount: 0 }

people> db.person.deleteMany({})
    acknowledged: true, deletedCount: 3 }

people> db.samples.insertOne({_id:1, name:"Bob", age:23})
    acknowledged: true, insertedId: 1 }

people> db.samples.insertOne({_id:1, name:"Tom", age:29})
MongoServerError: E11000 duplicate key error collection: people.samples inde
x: _id_ dup key: { _id: 1 }
people>

```

```

MongoServerError: E11000 duplicate key error collection: people.samples index: _id_ dup key: { _id: 1 }
people> use contactData
switched to db contactData
contactData> db.persons.insertOne({name:"Henry", age:26, hobbies:["Sports",
"Cooking"]})
{
  acknowledged: true,
  insertedId: ObjectId('6aa43483f99287b5cee5545d')
}
contactData> db.persons.insertOne({name:"Manuel", age:31, hobbies:["Cooking",
"Reading"]})
{
  acknowledged: true,
  insertedId: ObjectId('6aa43496f99287b5cee5545e')
}
contactData> db.persons.insertMany([
|   {name:"Bob", age:29, hobbies:["Hiking", "Yoga"]},
|   {name:"Maria", age:21},
|   {name:"Samuel", age:34}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aa434a1f99287b5cee5545f'),
    '1': ObjectId('6aa434a1f99287b5cee55460'),
    '2': ObjectId('6aa434a1f99287b5cee55461')
  }
}
contactData> |

```

```
mongosh mongodb://127.0.0.1:27027/
contactData> db.persons.find().pretty()
[
  {
    _id: ObjectId('6aa43483f99287b5cee5545d'),
    name: 'Henry',
    age: 26,
    hobbies: [ 'Sports', 'Cooking' ]
  },
  {
    _id: ObjectId('6aa43496f99287b5cee5545e'),
    name: 'Manuel',
    age: 31,
    hobbies: [ 'Cooking', 'Reading' ]
  },
  {
    _id: ObjectId('6aa434a1f99287b5cee5545f'),
    name: 'Bob',
    age: 29,
    hobbies: [ 'Hiking', 'Yoga' ]
  },
  { _id: ObjectId('6aa434a1f99287b5cee55460'), name: 'Maria', age: 21 },
  { _id: ObjectId('6aa434a1f99287b5cee55461'), name: 'Samuel', age: 34 },
  { _id: ObjectId('6aa434bbf99287b5cee55462'), name: 'Phil', age: 35 },
  { _id: ObjectId('6aa434c6f99287b5cee55463'), name: 'Kim', age: 24 },
  { _id: ObjectId('6aa434c6f99287b5cee55464'), name: 'Hans', age: 38 },
  { _id: ObjectId('6aa435acf99287b5cee55465'), name: 'Chrissy', age: 41 }
]
contactData> db.persons.find({age:{$gte:30}}).pretty()
[
  {
    _id: ObjectId('6aa43496f99287b5cee5545e'),
    name: 'Manuel',
    age: 31,
    hobbies: [ 'Cooking', 'Reading' ]
  },
  { _id: ObjectId('6aa434a1f99287b5cee55461'), name: 'Samuel', age: 34 },
  { _id: ObjectId('6aa434bbf99287b5cee55462'), name: 'Phil', age: 35 },
  { _id: ObjectId('6aa434c6f99287b5cee55464'), name: 'Hans', age: 38 },
  { _id: ObjectId('6aa435acf99287b5cee55465'), name: 'Chrissy', age: 41 }
]
contactData> |
```

```
mongosh mongodb://127.0.0.1 × + ▾
{ _id: 'cooking', name: 'Cooking' },
{ _id: 'yoga', name: 'Yoga' }

contactData> db.hobbies.insertMany([
  { _id: "reading", name: "Reading" },
  { _id: "cooking", name: "Cooking" },
  { _id: "hiking", name: "Hiking" }
], {ordered: false})
Uncaught:
MongoBulkWriteError: E11000 duplicate key error collection: contactData.hobbies
  index: _id_ dup key: { _id: "cooking" }
  result: BulkWriteResult {
    insertedCount: 2,
    matchedCount: 0,
    modifiedCount: 0,
    deletedCount: 0,
    upsertedCount: 0,
    upsertedIds: {},
    insertedIds: { '0': 'reading', '2': 'hiking' }
  }
  writeErrors: [
    WriteError {
      err: {
        index: 1,
        code: 11000,
        errmsg: 'E11000 duplicate key error collection: contactData.hobbies in
ex: _id_ dup key: { _id: "cooking" }',
        errInfo: undefined,
        op: { _id: 'cooking', name: 'Cooking' }
      }
    }
  ]

contactData> db.hobbies.find().pretty()

{ _id: 'sports', name: 'Sports' },
{ _id: 'cooking', name: 'Cooking' },
{ _id: 'yoga', name: 'Yoga' },
{ _id: 'reading', name: 'Reading' },
{ _id: 'hiking', name: 'Hiking' }

contactData> |
```

```
mongosh mongodb://127.0.0.1:27027/
contactData> db.movies.find({runtime:{$eq:30}}).pretty()
contactData> db.movies.find({runtime:{$gte:60}}).pretty()
contactData> db.movies.find({runtime:{$ne:60}}).pretty()
contactData> db.movies.find({"rating.average":{$gt:7}}).pretty()
contactData> db.movies.find({genres:"drama"}).pretty()
contactData> db.movies.find({genres:["Drama"]}).pretty()
contactData> db.movies.find({runtime:{$in: [30, 42]}}).pretty()
contactData> db.movies.find({runtime:{$nin: [30, 42]}}).pretty()
contactData> db.movies.find({$or: [{"rating.average":{$lt:5}}, {"rating.average":{$gt:9}]}).pretty()
contactData> db.movies.find({$nor: [{"rating.average":{$lt:5}}, {"rating.average":{$gt:9}]}).pretty()
contactData> db.movies.find({$and: [{"rating.average":{$gt:9}}, {genres:"Drama"}]}).pretty()
contactData> db.movies.find({"rating.average":{$gt:9}, genres:"Drama"}).pretty()
contactData> db.movies.find({$and: [{genres: "Drama"}, {genres: "Horror"}]}).count()
0
contactData> db.movies.find({runtime:{$not:{$eq:60}}}).count()
0
contactData> db.movies.find({runtime:{$ne:60}}).count()
0
contactData> use user
switched to db user
user> db.users.insertMany([{"name":"Max", hobbies: [{"title":"Sports", frequency:3}, {"title":"Cooking", frequency:6}], phone:475489575}, {"name":"Manuel", hobbies: [{"title":"Cooking", frequency:4}, {"title":"Cars", frequency:2}], phone:73874858459, age:39}])
```

```

count()
)
contactData> db.movies.find({runtime:{$not:{$eq:60}}}).count()
)
contactData> db.movies.find({runtime:{$ne:60}}).count()
)
contactData> use user
switched to db user
user> db.users.insertMany([{name:"Max", hobbies: [{title:"Sports", frequency
3},{title:"Cooking", frequency:6}], phone:475489575}, {name:"Manuel", hobbi
s: [{title:"Cooking", frequency:4}, {title:"Cars", frequency:2}], phone:738
74858459, age:39}])

  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aa43869f99287b5cee55466'),
    '1': ObjectId('6aa43869f99287b5cee55467')
  }
}

user> db.users.find().pretty()

{
  _id: ObjectId('6aa43869f99287b5cee55466'),
  name: 'Max',
  hobbies: [
    { title: 'Sports', frequency: 3 },
    { title: 'Cooking', frequency: 6 }
  ],
  phone: 475489575
},
{
  _id: ObjectId('6aa43869f99287b5cee55467'),
  name: 'Manuel',
  hobbies: [ { title: 'Cooking', frequency: 4 }, { title: 'Cars', frequenc
y: 2 } ],
  phone: 73874858459,
  age: 39
}

user> |

```

```

mongosh mongodb://127.0.0.1:27027/ > use test
test> db.users.insertMany([
  {
    _id: ObjectId('6aa43869f99287b5cee55467'),
    name: 'Manuel',
    hobbies: [ { title: 'Cooking', frequency: 4 }, { title: 'Cars', frequency: 2 } ],
    phone: '73874858459',
    age: 39
  },
  {
    _id: ObjectId('6aa438abf99287b5cee55468'),
    name: 'Anna',
    hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency: 3 } ],
    phone: '9758948574',
    age: null
  }
])
test> db.users.find({age: {$exists: false}}).pretty()
[
  {
    _id: ObjectId('6aa43869f99287b5cee55466'),
    name: 'Max',
    hobbies: [
      { title: 'Sports', frequency: 3 },
      { title: 'Cooking', frequency: 6 }
    ],
    phone: '475489575'
  }
]
test> db.users.find({age: {$exists: true, $ne: null}}).pretty()
[
  {
    _id: ObjectId('6aa43869f99287b5cee55467'),
    name: 'Manuel',
    hobbies: [ { title: 'Cooking', frequency: 4 }, { title: 'Cars', frequency: 2 } ],
    phone: '73874858459',
    age: 39
  }
]
test>

```

```
mongosh mongodb://127.0.0.1:27021/ > use users
MongoDB shell version: 4.0.1
connect: success
switched to db users
> db.users.findOne()
{
  name: 'Trump',
  hobbies: [
    { title: 'Coding', frequency: 2 },
    { title: 'Hiking', frequency: 3 }
  ],
  phone: '9758948574',
  age: null
}

> db.users.find({"hobbies.title":"Sports"}).pretty()
{
  _id: ObjectId('6aa43869f99287b5cee55466'),
  name: 'Max',
  hobbies: [
    { title: 'Sports', frequency: 3 },
    { title: 'Cooking', frequency: 6 }
  ],
  phone: 475489575
},
{
  _id: ObjectId('6aa438abf99287b5cee55468'),
  name: 'Anna',
  hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency: 3 } ],
  phone: 9758948574,
  age: null
}

> db.users.find({hobbies:{title:"Sports",frequency:2}}).pretty()
{
  _id: ObjectId('6aa438abf99287b5cee55468'),
  name: 'Anna',
  hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency: 3 } ],
  phone: 9758948574,
  age: null
}

user> |
```

```

mongosh mongodb://127.0.0.1:27020/
> use users
switched to db users
> db.users.findOne()
{
  phone: 9758948574,
  age: null
}
]
user> db.users.find({hobbies:{title:"Sports",frequency:2}}).pretty()
[
  {
    _id: ObjectId('6aa438abf99287b5cee55468'),
    name: 'Anna',
    hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency
: 3 } ],
    phone: 9758948574,
    age: null
  }
]
user> db.users.find({hobbies:{$size:3}}).pretty()

user> db.users.find({hobbies:{$all:["Sports","Cooking"]}}).pretty()

user> db.users.find({$and:[{"hobbies.title":"Sports"}, {"hobbies.frequency":{
$gte:2}}]}).pretty()
[
  {
    _id: ObjectId('6aa43869f99287b5cee55466'),
    name: 'Max',
    hobbies: [
      { title: 'Sports', frequency: 3 },
      { title: 'Cooking', frequency: 6 }
    ],
    phone: 475489575
  },
  {
    _id: ObjectId('6aa438abf99287b5cee55468'),
    name: 'Anna',
    hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency
: 3 } ],
    phone: 9758948574,
    age: null
  }
]
user>

```

```
mongosh mongodb://127.0.0.1:27021/
user> db.users.find({hobbies:{$size:3}}).pretty()

user> db.users.find({hobbies:{$all:["Sports","Cooking"]}}).pretty()

user> db.users.find({$and:[{"hobbies.title":"Sports"}, {"hobbies.frequency":{
$gte:2}}]}).pretty()
[
  {
    _id: ObjectId('6aa43869f99287b5cee55466'),
    name: 'Max',
    hobbies: [
      { title: 'Sports', frequency: 3 },
      { title: 'Cooking', frequency: 6 }
    ],
    phone: 475489575
  },
  {
    _id: ObjectId('6aa438abf99287b5cee55468'),
    name: 'Anna',
    hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency
: 3 } ],
    phone: 9758948574,
    age: null
  }
]
user> db.users.find({hobbies: {$elemMatch: {title:"Sports", frequency:{$gte:
3}}}}).pretty()
[
  {
    _id: ObjectId('6aa43869f99287b5cee55466'),
    name: 'Max',
    hobbies: [
      { title: 'Sports', frequency: 3 },
      { title: 'Cooking', frequency: 6 }
    ],
    phone: 475489575
  }
]
user> use movie
switched to db movie
movie>
```

```
mongosh mongodb://127.0.0.1:27027/
movie> db.movies.find().sort({"rating.average":1}).skip(239).limit(10).pretty()

movie> db.movies.find({}, {name:1, genres:1, runtime:1, rating:1}).pretty()

movie> db.movies.find({genres: "Drama"}, {"genres.$": 1}).pretty()

movie> db.movies.find({genres:{$all:["Drama","Horror"]}}, {"genres.$": 1}).pretty()

movie> db.movies.find({genres:"Drama"}, {genres:{$elemMatch:{$eq:"Horror"}}}).pretty()

movie> db.movies.find({"rating.average":{$gt:9}}, {genres:{$elemMatch:{$eq:"Horror"}}}).pretty()

movie> db.movies.find({"rating.average":{$gt:9}}, {genres:{$slice:2}, name:1}).pretty()

movie> db.movies.find({"rating.average":{$gt:9}}, {genres: {$slice:[1,2]}, name:1}).pretty()

movie> use user
switched to db user
user> db.users.updateOne({name:"Trump"},{$set: {age: 78}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
user> db.users.updateOne({name:"Trump"},{$inc: {age:1}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
user>
```

```
mongosh mongodb://127.0.0.1:27017/ > use test
test> db.users.find().pretty()
[
  {
    _id: ObjectId('6aa43869f99287b5cee55466'),
    name: 'Max',
    hobbies: [
      { title: 'Sports', frequency: 3 },
      { title: 'Cooking', frequency: 6 }
    ],
    phone: 475489575
  },
  {
    _id: ObjectId('6aa43869f99287b5cee55467'),
    name: 'Manuel',
    hobbies: [ { title: 'Cooking', frequency: 4 }, { title: 'Cars', frequency: 2 } ],
    phone: 73874858459,
    age: 39
  },
  {
    _id: ObjectId('6aa438abf99287b5cee55468'),
    name: 'Anna',
    hobbies: [ { title: 'Sports', frequency: 2 }, { title: 'Yoga', frequency: 3 } ],
    phone: 9758948574,
    age: null
  },
  {
    _id: ObjectId('6aa4393bf99287b5cee55469'),
    name: 'Trump',
    hobbies: [
      { title: 'Coding', frequency: 2 },
      { title: 'Hiking', frequency: 3 }
    ],
    phone: '9758948574',
    age: 80,
    isSporty: false
  }
]
```